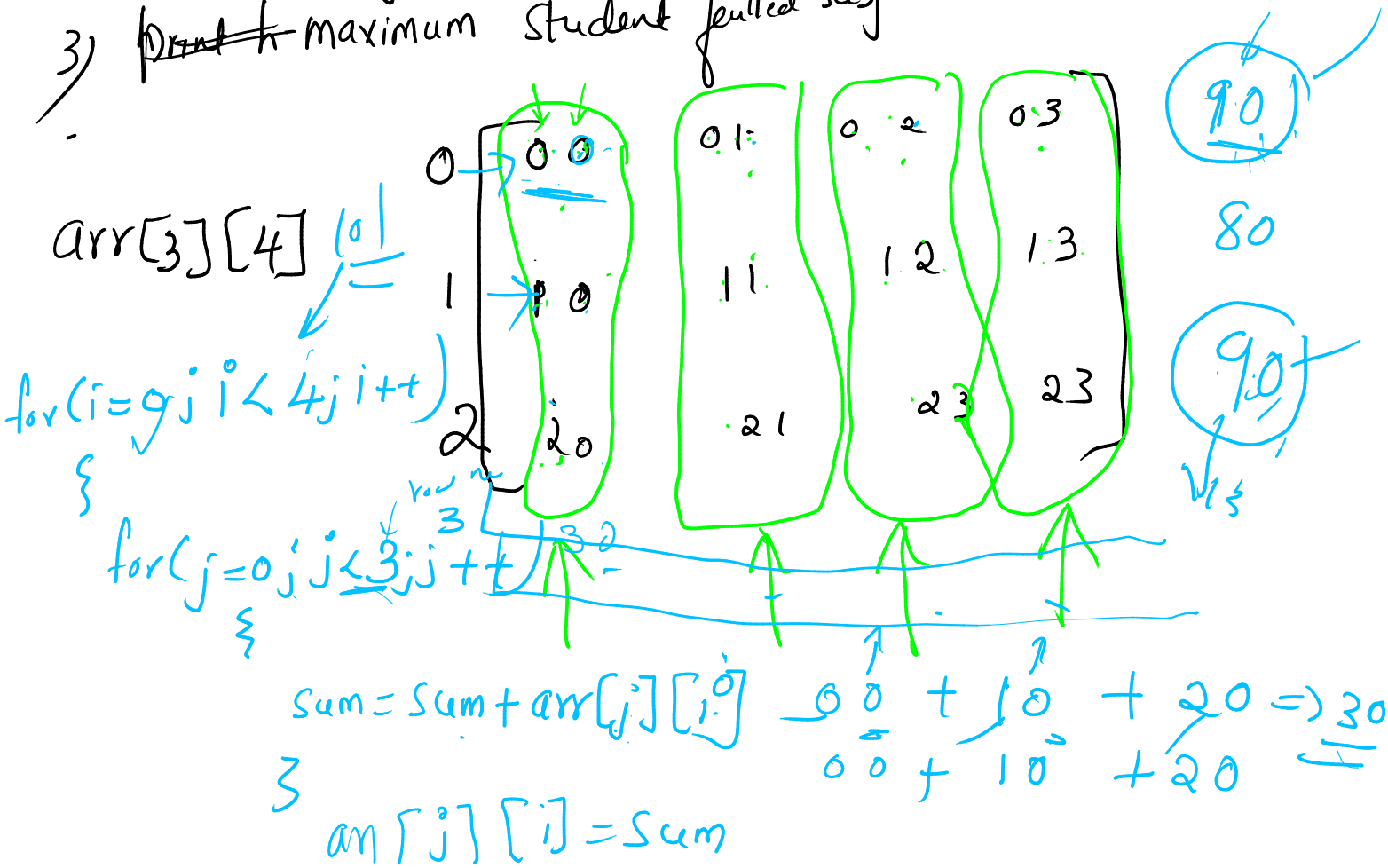
1) Total marks

2) who scores highest and highest marks

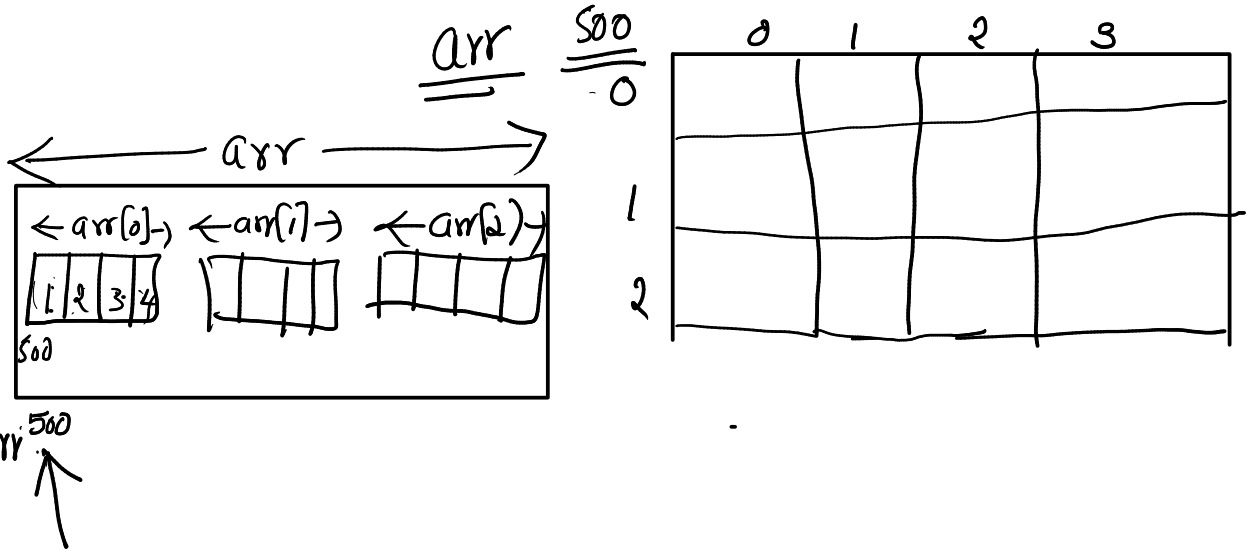
3) ~~Print~~ maximum Student failed subject



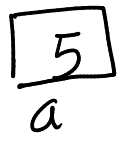
Q: How do you pass 2D array to a function?

datatype arrayname[row][col];

int arr[3][4];

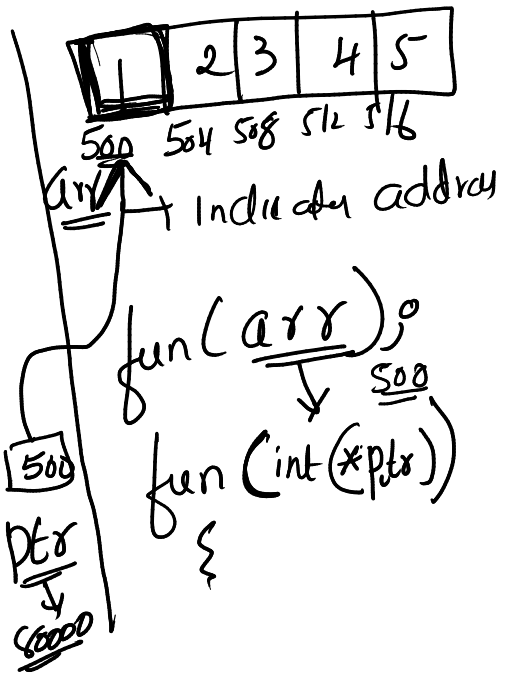


```
int a = 5;
fun(a);
```




```
fun(int x)
{
```





2D-array => arr[3][4]

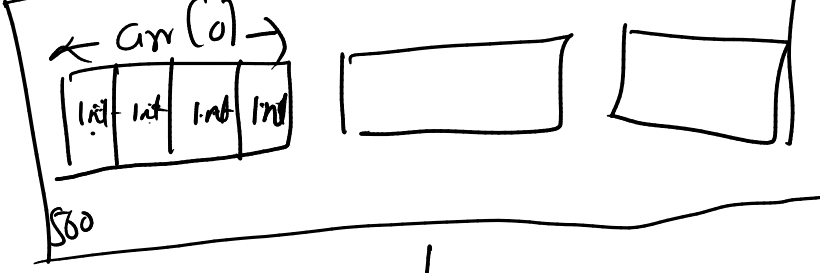
→ addrs = 500

fun(arr);

fun(500);

→ address passing

pointer needed



500 col

fun(int (*ptr)[4])

datatype (*pointername)[size]

col;

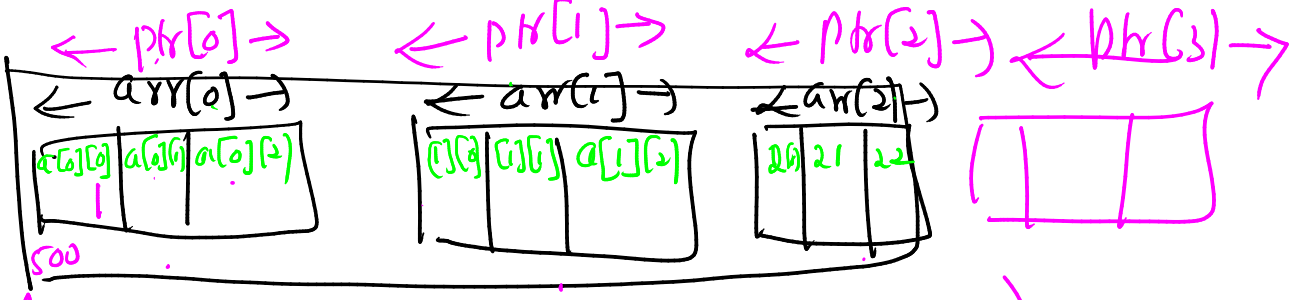
2D-array passing => pointer to an array.

int arr[3][5];

fun(arr);

→ col

int (*ptr)[5];

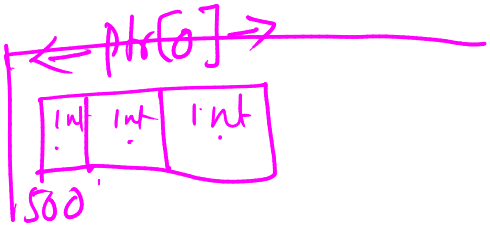


500
ptr

for(i=0; i<4; i++)
{
for(j=0; j<3; j++)
{
scanf("%d", &ptr[i][j]);
}

i=0 fixed, varying
ptr[0][0] ← j=0
ptr[0][1] ← j=1 ✓
ptr[0][2] ← j=2

j<3 ✗



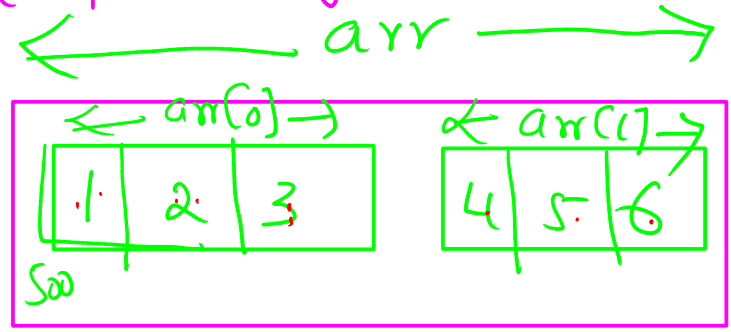
int (*ptr)[3];

↳ sizeof(ptr) ⇒ 500

↳ sizeof(*ptr) ⇒ 3 × 4 bytes ⇒ 12 bytes

↳ sizeof(**ptr) ⇒ 4 bytes.

int arr[2][3];
row, col



sizeof(arr) ⇒ 2D array ⇒ 24 bytes

sizeof(*arr) ⇒ 1D array ⇒ 12 bytes

arr
500

$$rows = \frac{sizeof(arr)}{sizeof(*arr)} \Rightarrow \frac{24}{12} \Rightarrow 2 \text{ rows}$$

sizeof(*arr) => 10 => 12 bytes

sizeof(**arr) => int => data element => 4 bytes

col = sizeof(*arr) // 1D array / sizeof(**arr) // element = 12/4 => 3

1) WAP to add two matrices and store it in a new array?

int arr[2][3], b[2][3], c[2][3];

1) input(arr, 2, 3);
2) input(barr, 2, 3);
3) print(arr, 2, 3);
4) print(barr, 2, 3);

arr[2][3] = [[1, 2, 1], [1, 1, 3]] (2x3)

b[2][3] = [[1, 2, 1], [4, 2, 1]] (2x3)

c[2][3]

arr[2][3]:

	00	01 ✓	02
0	1	2	1
1	10	11	12
	1	1	3

barr[2][3]:

	00	01	02
0	1	2	1
1	10	11	12
	4	2	1

2	4	2
5	3	4

c[2][3] =>

	00	01	02
0	00+00	01+01	02+02
1	10	11	12
	10+10	11+11	12+12

```
for(i=0; i<2; i++)
{
  for(j=0; j<3; j++)
  {
    c[i][j] = arr[i][j] + barr[i][j];
  }
}
```

i = 0 , j = 0, 1, 2
i = 1 , j = 0, 1, 2

c[2][3] =

	00	01	02
0	2	4	6
1	10	11	12
	5	3	4

i=0, 0<2 ✓
j: j=0, 1, 2, 3✗

c[0][0] = arr[0][0] + barr[0][0]
c[0][0] = 1 + 1

j = 1

c[i][j] = arr[i][j] + barr[i][j]
c[0][1] = arr[0][1] + barr[0][1]
c[0][1] = 2 + 2 => 4

j = 2; 2<3

c[i][j] = arr[i][j] + barr[i][j]
c[0][2] = arr[0][2] + barr[0][2]
c[0][2] = 1 + 1 => 2

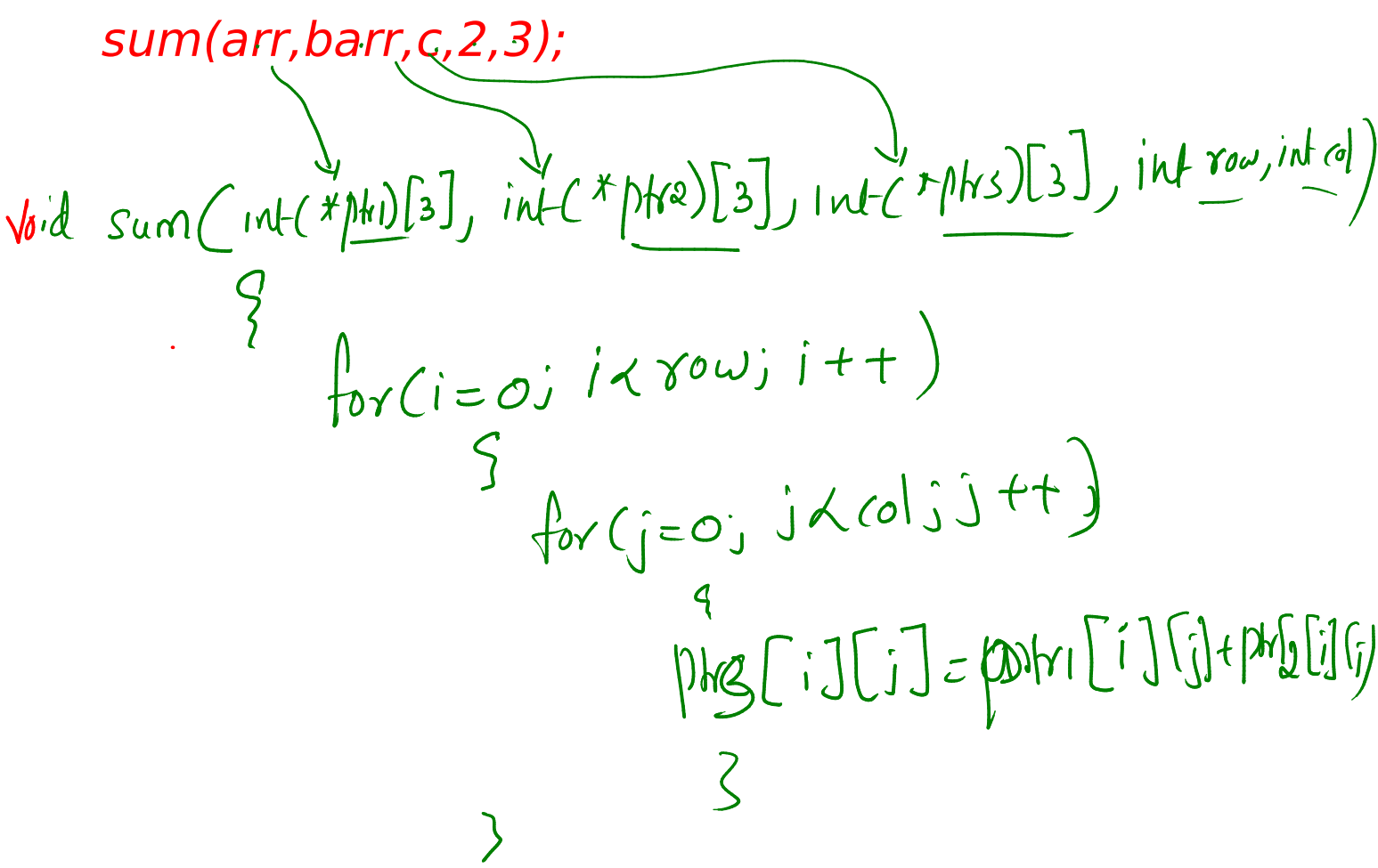
i = 1, 1<2 ✓
innerloop => j = 0, 1, 2, 3✗

c[i][j] = arr[i][j] + barr[i][j]
≠ arr[i][0] + barr[1][0]
c[1][0] = 1 + 4 = 5

j = 1, j<3

c[i][j] = arr[i][j] + barr[i][j]
c[1][1] = arr[1][1] + barr[1][1]
c[1][1] = 1 + 2 => 3

i = 1, j = 2, 2<3
c[i][j] = arr[i][j] + barr[i][j]
c[1][2] = arr[1][2] + barr[1][2]
c[1][2] = 3 + 1 => 4


sum(arr,barr,c,2,3);
void sum(int(*ptr1)[3], int(*ptr2)[3], int(*ptr3)[3], int row, int col)
{
for(i=0; i<row; i++)
{
for(j=0; j<col; j++)
{
ptr3[i][j] = ptr1[i][j] + ptr2[i][j]
}
}